

Stage-3 Raw CDF Refinement Distillation, Ablation, and Final Metrics

Mie Spray Postprocessing Project

2026 年 5 月 10 日

Claim: Stage-3 refinement turns the Stage-2 heteroscedastic teacher into a raw-CDF-aware predictor by mixing direct raw supervision with teacher distillation under time-dependent coverage regimes.

Current status after the $\Delta P^{0.5}$ feature-engineering update:

- the teacher/student feature space uses $A_{\text{scale}} = \Delta P^{0.5} \rho_a^{-0.25} d_n^{0.5}$;
- Stage-3 is evaluated over five seeds in the mode-C pipeline;
- the dominant winning ablation is `anchor_off`;
- external clean evaluation has been completed on 795 561 points and 33 845 trajectories;
- empirical baselines and a sparse heteroscedastic GP are now included for comparison.

Why Stage-3 Exists

Stage-2 teacher

Stage 2 learns a heteroscedastic Gaussian model from fitted representative curves. It is smooth and defined everywhere on 0–5 ms, but it can inherit fitting assumptions from the reduced curve model.

Raw CDF series

Raw CDF observations are closer to measured data, but they are incomplete and time-dependent: coverage thins as trajectories disappear from the field of view or fail quality filters.

Stage-3 refinement

Stage 3 uses raw CDF where it is reliable, teacher distillation where raw data are sparse, and light shape/onset priors to keep trajectories physically plausible.

CDF Coverage Regimes

Time is binned into $\Delta t = 0.1$ ms intervals. For each operating-condition group, smoothed CDF sample counts define three supervision regimes:

Regime	Trigger	Supervision role
raw_reliable	coverage $\geq 70\%$ of peak	raw CDF is trusted directly
raw_uncertain	below 70%, above 20%	teacher dominates transition
teacher_only	below 20% or < 4 samples	teacher is the only usable signal

Transitions require consecutive bins, which prevents isolated coverage dips from changing the regime.

Default Supervision Weights

Regime	Raw NLL weight	KD KL weight
raw_reliable	1.00	0.25
raw_uncertain	0.00	1.00
teacher_only	0.00	0.75

- Raw NLL is used only where raw CDF is reliable and finite.
- KD KL prevents the student from drifting where raw observations are sparse.
- A global inverse-frequency time-bin weight is clipped to 0.5–2.0.
- Teacher confidence further down-weights high- σ teacher regions.

Source: MLP/MLP_training/train_stage3_distillation_plus_raw_series.py

$\Delta P^{0.5}$ A-Scaled Physical Space

Current engineered scale:

$$A_{\text{scale}} = \Delta P^{0.5} \rho_a^{-0.25} d_n^{0.5}.$$

Network output is still scaled:

$$\hat{\mu}(t) = \mu(t)/A, \quad \hat{\sigma}(t) = \sigma(t)/A.$$

Stage-3 losses are computed in physical millimetres:

$$\mu_{\text{phys}} = A\hat{\mu}, \quad \sigma_{\text{phys}} = A\hat{\sigma}.$$

This lets raw CDF targets enter directly in mm while keeping the sparse pressure/diameter axes collapsed through the data-supported A -scale.

Source: MLP/MLP_training/engineered_feature_common.py

Composite Loss

$$\mathcal{L} = \mathcal{L}_{\text{raw}} + \mathcal{L}_{\text{KD}} + \lambda_{\text{onset}} \mathcal{L}_{\text{onset}} + \lambda_{\text{anchor}} \mathcal{L}_{\text{anchor}} + \lambda_{d_1} P_{d_1} + \lambda_{d_2} P_{d_2}.$$

Term	Role
Raw NLL	Gaussian NLL against raw CDF in physical space
KD KL	Gaussian KL between student and Stage-2 teacher in physical space
Onset BCE	soft ramp target in the first 0.2 ms
Anchor	optional early-time penalty on $\mu(t \approx 0)$
P_{d_1}	monotonicity penalty against negative increments
P_{d_2}	concavity penalty after $t \geq 0.5$ ms

Loss Terms: Probabilistic Meaning

Raw Gaussian NLL

Where raw CDF samples are reliable, the student is scored directly in physical mm:

$$\mathcal{L}_{\text{raw},i} = \frac{1}{2} \left[\frac{(y_i - \mu_i)^2}{\sigma_i^2} + \log \sigma_i^2 \right] + \text{const.}$$

This rewards both accurate means and correctly sized variances; shrinking σ_i is penalized if residuals remain large.

Teacher-student Gaussian KL

When raw coverage is weak, the student is kept close to the Stage-2 teacher:

$$D_{\text{KL}}(T||S) = \log \frac{\sigma_S}{\sigma_T} + \frac{\sigma_T^2 + (\mu_T - \mu_S)^2}{2\sigma_S^2} - \frac{1}{2}.$$

The direction $T||S$ treats the teacher distribution as the reference distribution that the student should cover in sparse regions.

Key Ablation Finding: Anchor Off

Old design assumption

The early anchor was originally included to keep $\mu(t \approx 0)$ close to zero and prevent unphysical onset behavior.

What the ablation found

The best current Stage-3 setting is usually `anchor_off`, i.e. $\lambda_{\text{anchor}} = 0$. In the five-seed $\Delta P^{0.5}$ mode-C pipeline, seeds 42, 17, 99, and 7 selected `anchor_off`; seed 2024 selected `raw_reliable_no_kd`.

Interpretation

The learned onset head, raw/KD regime weights, and shape priors are enough to regularize early time. A hard $\mu(0)$ anchor can overconstrain the raw-CDF refinement.

Metric Contract for Stage-3 Results

Accuracy metrics

RMSE emphasizes large misses; MAE reports the typical absolute millimetre error; P95 absolute error reports the tail. Bias keeps the sign, so it exposes systematic over- or under-prediction that RMSE/MAE hide.

Uncertainty metrics

Coverage is

$$\Pr(|y - \mu| \leq k\sigma), \quad k \in \{1, 2\}.$$

For a calibrated Gaussian, the nominal targets are 0.683 and 0.954. Higher values mean conservative intervals; lower values mean under-coverage.

Protocol separation

The five-seed in-pipeline result measures training stability on each seed's post-evaluation subset. The external clean diagnostic is the headline comparison because every model is evaluated on the same 795 561 points.

Five-Seed In-Pipeline Result

This is the mode-C pipeline bootstrap on the common post-evaluation test subset.

Metric	Mean $\pm$ std	95% CI
RMSE mm	8.924 $\pm$ 1.133	8.070–9.855
MAE mm	7.207 $\pm$ 1.000	6.507–8.076
Bias mm	-5.435 $\pm$ 1.655	-6.945–4.421
P95 abs. err. mm	17.114 $\pm$ 1.993	15.528–18.700
1 σ coverage	0.654 $\pm$ 0.067	0.601–0.706
2 σ coverage	0.966 $\pm$ 0.016	0.954–0.979

Important protocol note: this in-pipeline summary has $n = 8398$ post-evaluation points per seed. It is a stability diagnostic, not the external 795 561-point clean benchmark.

Source: MLP/runs_mlp/full_pipeline_C_20260509_110100/bootstrap_summary.json

External Clean Diagnostic

The same trained Stage-3 models were evaluated on the full clean diagnostic protocol: 795 561 points and 33 845 trajectories.

Model	RMSE	MAE	P95	1σ cov.
Hiroyasu–Arai calibrated	9.974	7.686	20.493	0.536
Naber–Siebers delay	8.840	6.780	17.922	0.620
Sparse heteroscedastic GP mean	6.898	5.206	14.365	0.670
Stage-3 MLP $\Delta P^{0.5}$, 5-seed mean	6.211	4.587	13.029	0.774

The Stage-3 MLP beats the GP on accuracy, while the GP remains closer to nominal uncertainty coverage.

Source: `Thesis/generated/baseline_comparison_20260509/headline_comparison.csv`

Clean vs Hard-Subset Calibration

External clean diagnostic

The $\Delta P^{0.5}$ Stage-3 MLP over-covers on the clean full diagnostic: $1\sigma/2\sigma$ coverage is 0.774/0.981, above nominal 0.683/0.954.

In-pipeline post-evaluation subset

The same pipeline has $1\sigma/2\sigma$ coverage of 0.654/0.966, much closer to nominal and slightly under at 1σ .

Paper interpretation

The learned σ is not a single global error bar. It behaves differently on clean/easy trajectories and harder representative rows, which is evidence for difficulty-aware uncertainty rather than pure input-independent noise.

What Should Go Into the Thesis

Method sentence

Stage 3 refines the Stage-2 heteroscedastic teacher against raw CDF measurements using regime-dependent raw NLL and teacher KL losses, with regimes determined by per-condition CDF coverage over time.

Ablation sentence

The best $\Delta P^{0.5}$ Stage-3 configuration removes the explicit early-time anchor in four of five seeds, indicating that the onset head and raw/KD regime weighting provide sufficient early-time regularization.

Result sentence

On the external clean benchmark, the resulting five-seed Stage-3 MLP reaches RMSE 6.211 ± 0.283 mm and outperforms Hiroyasu–Arai, Naber–Siebers, and a sparse heteroscedastic GP on RMSE, MAE, and P95 error.

Data Sources for Thesis Extraction

Item	Path
Stage-3 training script	MLP/MLP_training/train_stage3_distillation_plus_raw_series.py
Five-seed pipeline summary	MLP/runs_mlp/full_pipeline_C_20260509_110100/bootstrap_summary.json
Representative winner run	MLP/runs_mlp/distill_cdf_onset_v2_ablate_anchor_off_20260509_111556/
External comparison table	Thesis/generated/baseline_comparison_20260509/headline_comparison.csv
GP baseline output	MLP/runs_mlp/gp_baseline_20260509_201444/